Practical Report: CI/CD Pipeline for quakeloc

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1. Project and links

quakeloc detects earthquakes (STA/LTA), picks P-wave onsets (AIC) and locates epicentres with 95 % error ellipses (Geiger's method). It is a FastAPI service validated on synthetic data and on a frozen snapshot of the KNET network near Bishkek. Repository: https://github.com/a169n/applied_5. Pipeline: Jenkinsfile. Image: https://hub.docker.com/r/a169n/quakeloc. Steps to reproduce: docs/RUNBOOK.md.

2. Steps

Docker. A multi-stage `Dockerfile` has separate `test`, `artifacts`, `data` and `runtime` targets. The runtime image contains no compilers or test tools and runs as a non-root user. It is 516 MB locally and 108.9 MB compressed on Docker Hub (Figure 3). A build fails if the calibration artifacts it bakes in differ from those the tests validated.

Jenkins. Jenkins runs in Docker and is configured entirely from the repository (Configuration as Code and Job DSL), so it can be recreated with one command. A multibranch job scans GitHub every 2 minutes, because a local Jenkins cannot receive webhooks. The ten stages are checkout, test image, lint, unit and property tests, scientific validation, real-data validation, runtime image, container smoke test, Docker Hub push and Minikube deploy. The last two run only on `main`. A separate job refreshes the data and never pushes; its first run fetched files identical to the committed snapshot.

Git and automation. Work reached `main` through a pull request, and every build and test stage has a `make` target with the same command, so CI failures can be reproduced locally.

Deployment (bonus). Stage 10 deploys the image tagged with the commit SHA to Minikube, runs a smoke Job inside the cluster and rolls back on failure. I tested the rollback manually.

3. Results

Green run. `main` build #1 passed all stages in 2 min 23 s (Figure 1) with 173 tests passing (Figure 4). Most stages were fast because the branch build had already filled the Docker layer cache. `/version` in the cluster returned the SHA of `main` HEAD, `d5ba3ca`.



Figure 1. Green pipeline on `main`, build #1.

Red run. Branch `demo/broken-locator` negates one column of the location Jacobian. Stage 4 failed with 47 of 162 tests failing, including `test_noise_free_exact_inversion` (Figure 2). Nothing was pushed or deployed: Docker Hub and the cluster stayed on `d5ba3ca`. This branch is never merged.

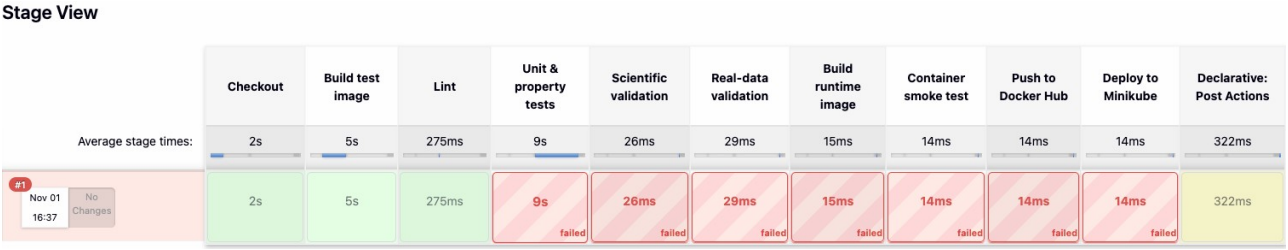


Figure 2. Red pipeline on `demo/broken-locator`. The console reports stages 5 to 10 as "skipped due to earlier failure"; Stage View paints them as failed.

Cluster state after the deploy (`kubectl -n quakeloc get pods`):

```
NAME                          READY   STATUS      RESTARTS   AGE
quakeloc-57945b5df7-pmx4k     1/1     Running     0          5m25s
quakeloc-57945b5df7-tzzvg     1/1     Running     0          5m41s
quakeloc-smoke-6gcrg          0/1     Completed   0          5m19s
```

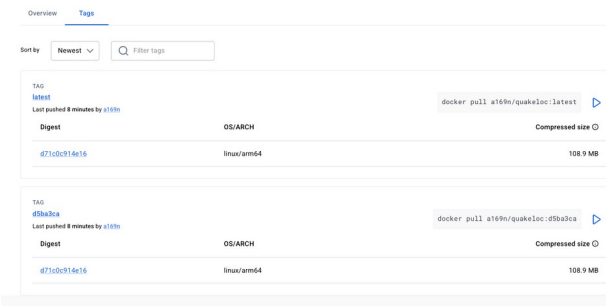


Figure 3. Docker Hub tags.

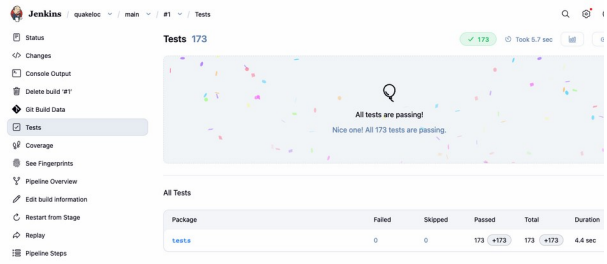


Figure 4. Jenkins test report.

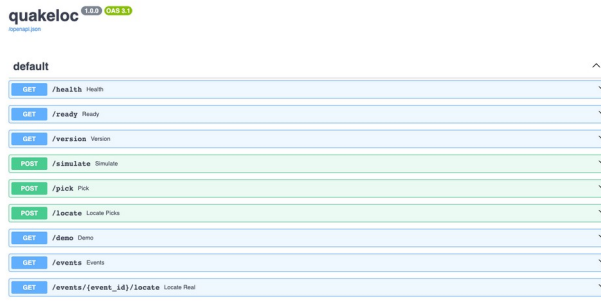


Figure 5. API documentation at `/docs`.

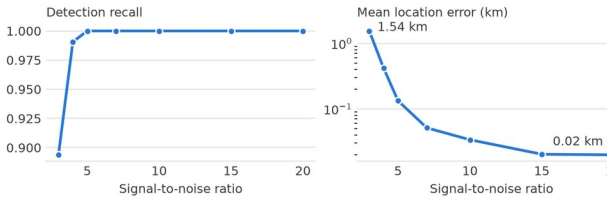


Figure 6. Synthetic SNR sweep, seed 42.

Pretty-print


```
{"id":"usb000r68x","located":
{"lat":42.48822684349082,"lon":74.7994544164084,"t0_s":-0.9427035441425258,"rms_s":0.12672387770929092},"ellipse":
{"a_km":1.679609892072253,"b_km":1.4258933430998413,"azimuth_deg":132.3182278936265},"usgs":
{"lat":42.5547,"lon":74.816,"depth_km":9.92,"magnitude":4.2,"time":"2014-05-
28T02:25:59.670000Z"},"mislocation_km":7.514824460787507,"used":
["AAK","AML","CHM","EKS2","KBK","KZA","TKM2","UCH","USP"],"rejected":{},"skipped":{"ULHL":"LTA is zero: dead segment in the
trace"}}
```


Figure 7. Response of `/events/usb000r68x/locate` from the Minikube deployment.

Science. On synthetic data, recall rises from 0.89 at SNR 3 to 1.0 at SNR 5 and above, and mean location error falls from 1.54 km to 0.02 km (Figure 6). On real data all 4 events locate, at 2.8, 7.5, 12.2 and 18.3 km from the USGS epicentres. Four events are a demonstration, not a statistical validation. The ellipses cover pick scatter only, so the USGS epicentre often lies outside them.

4. Difficulties

- **Sparse real data.** EarthScope returned no KN waveforms for much of the catalog period, so only 4 events qualified and lowering the magnitude limit to 3.5 added none. I kept the 4 events and stated the limit instead of tuning the results.
- **Stale artifacts.** The identity check failed locally after `src/` changed, because the artifacts were older than the code. Recalibrating fixed it, and the runbook now lists this case.
- **Jenkins lost its network.** Recreating the container dropped its link to the Minikube network, so `make jenkins-up` now restores it.
- **A test hid failures.** A stub's final `exit 0` masked simulated `kubectl` failures, so a commit went in with two failing tests. The next commit fixed the stub.
- **Git on macOS.** `git push` hung on a keychain prompt; pushing through the GitHub CLI's credential helper solved it.

5. Conclusions

The pipeline turns a merge into a tested, published and deployed service without manual steps, and the red run shows that a scientific bug stops before release. The most useful decision was to test against known answers, since a generic smoke test would not have caught a sign error in the Jacobian. The main limits are the small real-data sample, the one-layer velocity model and a Jenkins instance that runs only on a laptop.